PRO-MOTION:
Automatic Prosthetic Leg Suspension System

AUTHORS

Cassie Gallogly, cgallogl@ucsc.edu
Clara Almeida, cjalmeid@ucsc.edu
Jackson McIntyre, jcmcinty@ucsc.edu
Nathan Tran, ntran48@ucsc.edu

ADVISORS

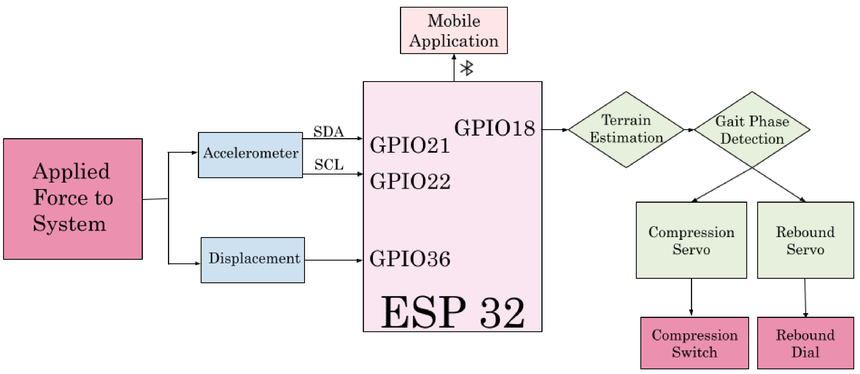
Professor Peterson
TA - Ravipa Losakul
TA - Kalean Schorger

INTRODUCTION

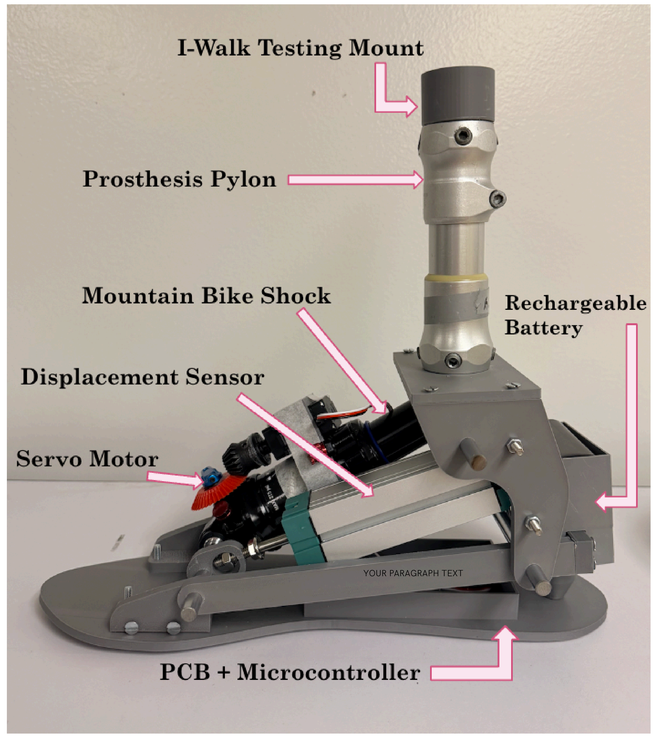
Lower-limb prosthesis users face significant challenges due to suspension systems that cannot adapt to changing activity or terrain. Current solutions often require amputees to own and switch between multiple prostheses, increasing financial and logistical difficulty.

PRO-MOTION addresses this gap by developing an all-terrain below-knee dynamically adjusting prosthetic suspension system that:

- Eliminates the need for multiple prostheses
- Automatically adjusts compression and rebound damping based on terrain and gait cycle phase
- Reports back user gait cycle analytics via a mobile application
- Allows for manual damping adjustment
- Meets the structural requirements for K-Level 4 ambulation



ANALYSIS



How Are We Tracking Gait?

- Each step causes a noticeable change in acceleration (Fig. 1).
- By measuring these acceleration thresholds we know when a user's foot makes contact with the ground, using our step detection algorithm (Fig.2)

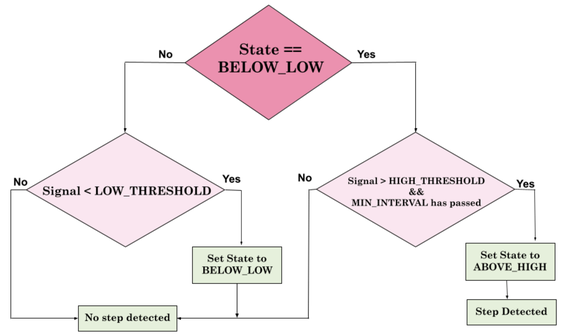


Fig. 2: Step detection algorithm

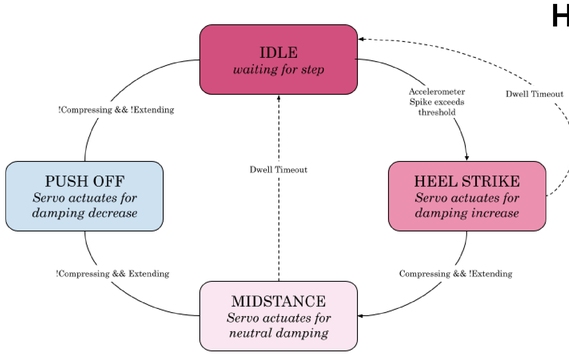
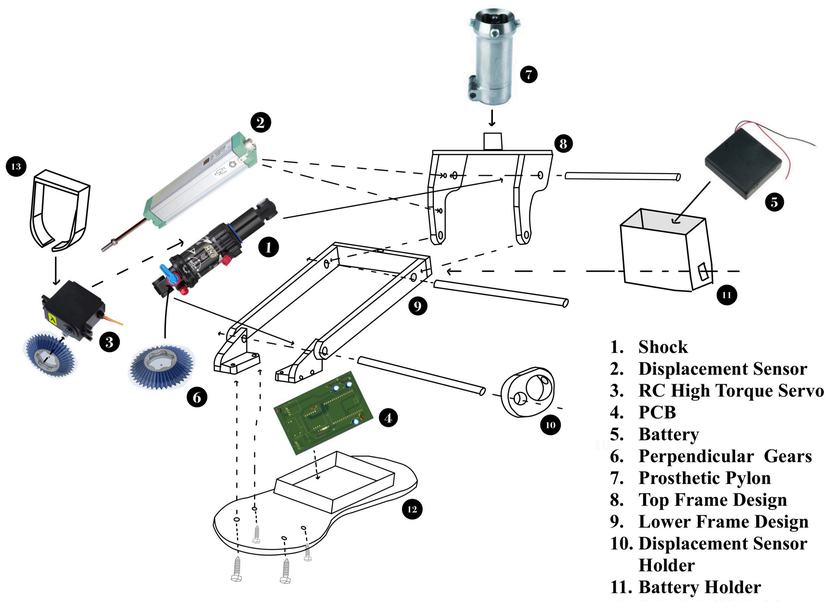


Fig. 3: Damping Control State Machine

How Are We Using This Data?

- User gait phase is determined by shock displacement and accelerometer data
- Prosthetic damping determined by gait phase/state (Fig. 3)
- Additional metrics collected for user gait health:
  - Cadence (step/minute)
  - Step-timing consistency

MECHANICAL INTEGRATION



1. Shock
2. Displacement Sensor
3. RC High Torque Servo
4. PCB
5. Battery
6. Perpendicular Gears
7. Prosthetic Pylon
8. Top Frame Design
9. Lower Frame Design
10. Displacement Sensor Holder
11. Battery Holder
12. Sole & PCB Holder
13. Servo Holder

CONCLUSION

Testing Approach

- 100-step immobilized knee gait trials to verify detection of impeded gait (Fig. 4)
- Cadence and step-time variability analysis
- Static load testing (target ≥100 lbs)
- Compression stroke and displacement sensor calibration
- Servo torque and RPM verification at rated voltage

Testing Results:

- Reliable step and impeded gait detection
- Shock meets target compression
- Displacement sensor achieves <3% error
- Servo delivers full actuation range

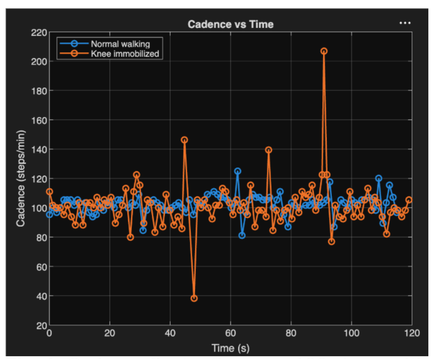


Fig. 4: Cadence fluctuation over 100 steps, with and without subject's knee immobilized

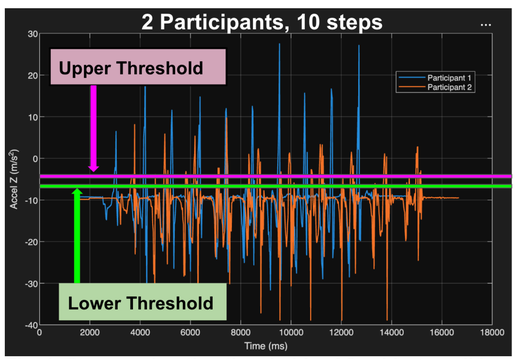


Fig. 1: Step acceleration thresholds of 2 subjects walking

PRO-MOTION demonstrated successful sensing and actuation for an adaptive prosthetic suspension system. Gait analysis confirmed that the system can distinguish normal from irregular gait via cadence variability.